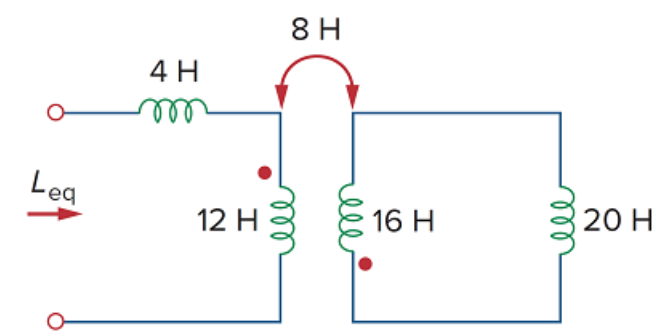


Problem

Determine the equivalent L_{eq} in the circuit of Fig. 13.81.

Figure 13.81



Step-by-step solution

Step 1 of 4

Refer to Figure 13.81 in the textbook.

First assume the frequency, $\omega = 1 \text{ rad/sec}$.

Calculate the impedances of the inductors.

$$Z_{4 \text{ H}} = j\omega L$$

$$= j(1)(4)$$

$$= j4 \Omega$$

$$Z_{12 \text{ H}} = j(1)(12)$$

$$= j12 \Omega$$

$$Z_{16 \text{ H}} = j(1)(16)$$

$$= j16 \Omega$$

$$Z_{20 \text{ H}} = j(1)(20)$$

$$= j20 \Omega$$

$$Z_{8 \text{ H}} = j(1)(8)$$

$$= j8 \Omega$$

Step 2 of 4

Apply 1 V voltage source at the input of the circuit and draw the circuit in frequency domain.

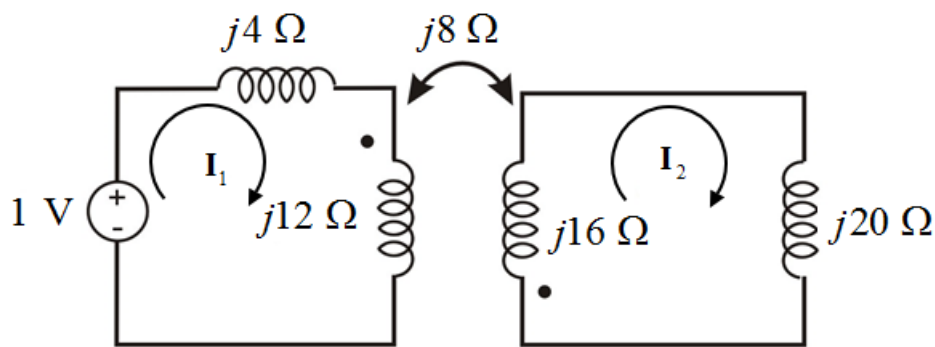


Figure 1

Step 3 of 4

Apply Kirchhoff's voltage law to loop1.

$$-1 + j4\mathbf{I}_1 + j8\mathbf{I}_2 + j12\mathbf{I}_1 = 0$$

$$j8\mathbf{I}_2 = 1 - j16\mathbf{I}_1$$

$$\mathbf{I}_2 = \frac{1}{j8} - \frac{j16\mathbf{I}_1}{j8}$$

$$\mathbf{I}_2 = -j0.125 - 2\mathbf{I}_1$$

Apply Kirchhoff's voltage law to loop2.

$$j20\mathbf{I}_2 + j16\mathbf{I}_2 + j8\mathbf{I}_1 = 0$$

$$j36\mathbf{I}_2 + j8\mathbf{I}_1 = 0$$

Substitute $-j0.125 - 2\mathbf{I}_1$ for $\mathbf{I}_2$.

$$j36(-j0.125 - 2\mathbf{I}_1) + j8\mathbf{I}_1 = 0$$

$$4.5 - j72\mathbf{I}_1 + j8\mathbf{I}_1 = 0$$

$$\mathbf{I}_1 = \frac{4.5}{j64}$$

Step 4 of 4

Calculate the equivalent impedance seen by the source.

$$\begin{aligned} \mathbf{Z}_{\text{eq}} &= \frac{\mathbf{V}_{\text{in}}}{\mathbf{I}_1} \\ &= \frac{1}{\left(\frac{4.5}{j64}\right)} \\ &= j14.22 \, \Omega \end{aligned}$$

Calculate the equivalent inductance seen by the source.

$$\begin{aligned} L_{\text{eq}} &= \frac{\mathbf{Z}_{\text{eq}}}{j\omega} \\ &= \frac{j14.22}{j(1)} \\ &= 14.22 \, \text{H} \end{aligned}$$

Thus, the equivalent inductance is 14.22 H.